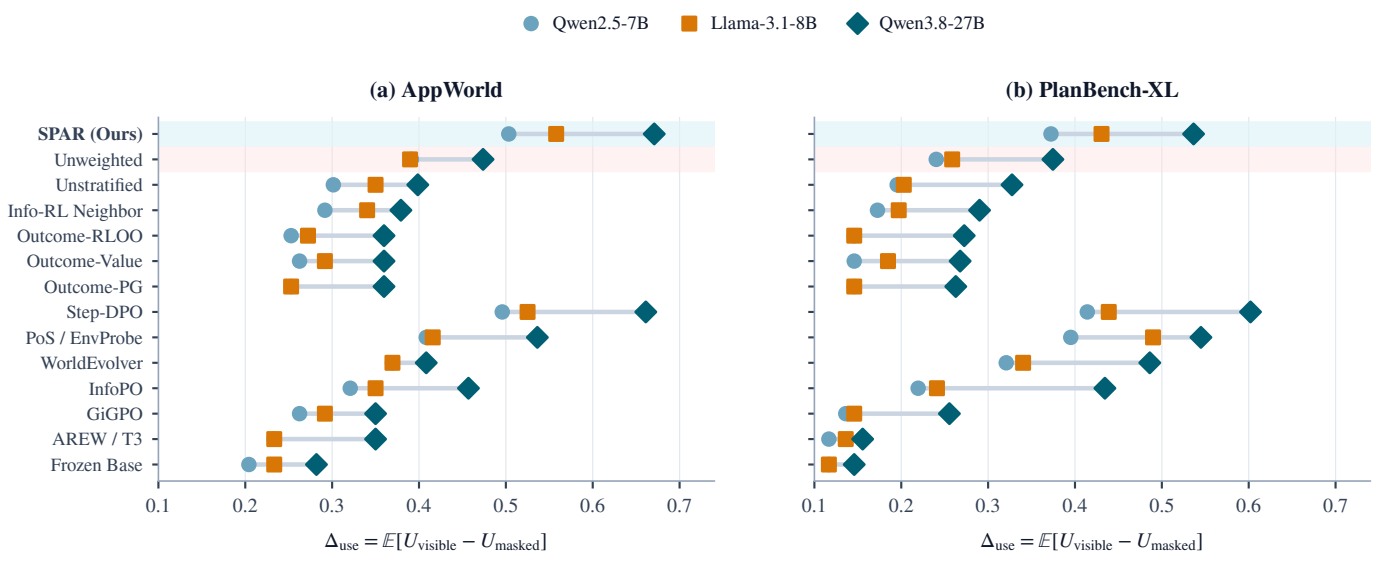
Qwen2.5-7B
Llama-3.1-8B
Qwen3.8-27B
(a) AppWorld
(b) PlanBench-XL
SPAR (Ours)
Unweighted
Unstratified
Info-RL Neighbor
Outcome-RLOO
Outcome-Value
Outcome-PG
Step-DPO
PoS / EnvProbe
WorldEvolver
InfoPO
GiGPO
AREW / T3
Frozen Base
0.1
0.2
0.3
0.4
0.5
0.6
0.7
$\Delta_{\text{use}} = \mathbb{E}[U_{\text{visible}} - U_{\text{masked}}]$